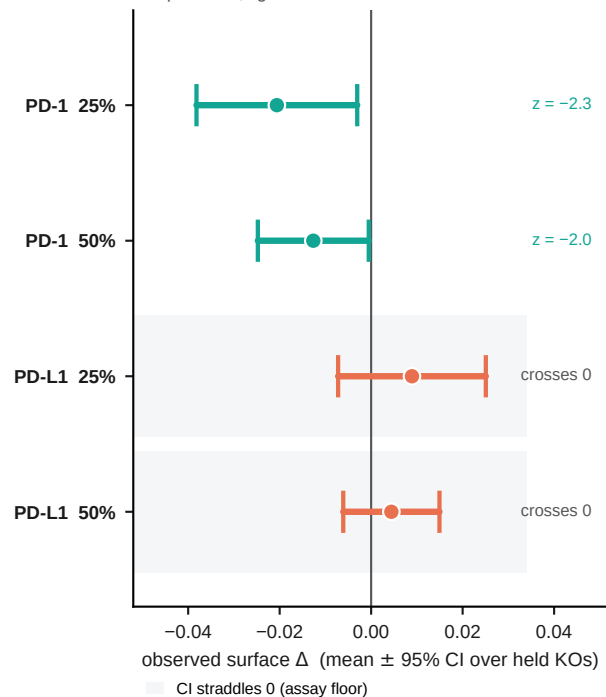


Second-dataset (n = 2) replication: checkpoint receptor vs ligand at the surface (Chen Perturb-icCITE-seq, primary human CD4+ Treg CRISPR-KO)

a

Checkpoint surface response (Chen)

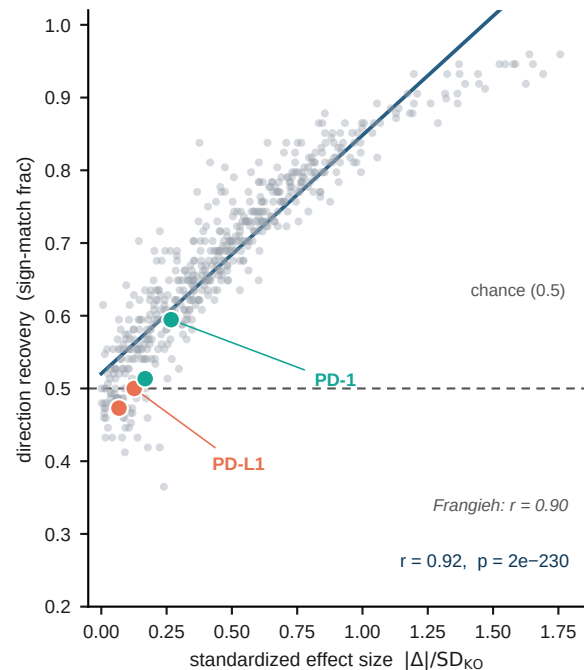
receptor off 0, ligand at the floor



b

Assay-floor confound replicates

277 surface markers $\times$ 2 holdouts



c

n = 2: asymmetry replicates

receptor > ligand in both; magnitude context-dependent

